

Effects of primary school's physical environment on children's spatial perception and behavior

The case of Kagithane, Istanbul, Turkey

Ahmet Türel and Elmira Ayşe Gür

Department of Architecture, Istanbul Technical University, Istanbul, Turkey

Primary
school's
physical
environment

425

Received 28 December 2018

Revised 20 February 2019

8 March 2019

17 April 2019

Accepted 1 May 2019

Abstract

Purpose – The relationship between the child and his/her physical environment is an area of interaction that includes social, psychological and cultural factors along with the spatial experience, perception and behavior of the child. This study is based on the effects of spatial perception and behavior of the child within the physical environment of primary schools. In this direction, the purpose of this paper is to investigate how spatial and physical characteristics of primary school typologies affect the spatial perception and behavior of the child. Also, the parameters affecting spatial perception and behavior are examined.

Design/methodology/approach – The question to be investigated is how the spatial and physical characteristics of the school's physical environment affect the child's spatial perception and behavior in primary schools with different typologies. Within this scope, Istanbul's Kagithane region is selected as a case study. Schools are chosen for their similar spatial and dimensional features and similar socio-economic environment. The methodology of the study consists of a literature review, an observational study carried out to discover the interaction between the child and his/her school building and the analysis of the student's cognitive maps. These maps were evaluated according to topological, projective, metric and imaginative parameters.

Findings – The results show spatial organization and physical characteristics of primary school buildings with a structure that allows for change and transformation, and contributes to the physical and cognitive development of children.

Originality/value – This study will provide an opportunity to develop the design of future primary school buildings that can support the spatial perception and spatial experiences of the children.

Keywords Spatial perception, Primary school, Cognitive development, Child perception, Cognitive map

Paper type Research paper

Introduction

Space can be defined as an important means of communication that perceives the environment and shapes behavior. Therefore, it provides opportunities for social and vital activities such as encountering, living, socializing and experiencing in society and human relations. For children of primary school age, school buildings become important as the first places they meet with different groups and social environments. The physical environment directly or indirectly influences the mechanism of perception and behavior. In this respect, the child needs a number of spaces to feel safe, to love the environments in which he/she lives and grows, to express himself/herself and to respond to his/her needs socially and culturally.

The study was based on the school's physical environment influencing the spatial perception and behavior of the child. These properties, such as characteristics of the primary school buildings, physical activity opportunities, accessibility, transportability, spatial organization, size-ratio-volume of the space, material, color and texture, are factors that affect the child's spatial perception and behavior. The aim of the study is to determine the effects of spatial and physical characteristics of the school environment on children's spatial perception and behavior in primary schools with different typologies. In this context, the study developed around the question of how the spatial and physical characteristics of



the school buildings in different typologies that are designed with or without context-specific data affect the spatial perception and behavior of children. Sub-research questions are in the following direction:

- the effects of school buildings that are designed with or without context-specific data on the spatial perception and cognitive development of children; and
- what the spatial and physical characteristics of the school buildings with different typologies that affect the spatial perception and behavior in primary school-aged children are and what the level of connection between the spatial perception of children and the use of space is.

In this context, a case study was conducted with literature research. Nef, Ziyapasa and Ismail Erez Primary Schools located in the Kagithane district of Istanbul were designated as a case. The names of the schools were used in the study with the required permits. Within the scope of the case study, observations were made in order to discover the interaction between the child and the school spaces. At the same time, a cognitive mapping study was conducted with the children. Within the scope of the research, the focus was on children between the ages of 6 and 11, who form the first four years of primary school education. This age group was chosen because of the rapid development of perceptual and cognitive development and the emergence of social relations.

In the study, the spatial organization of the school buildings' projects produced by the government and the growth of the child's development and spatial perception in the school building designed with context-specific data are to be discussed. In this respect, it is expected that the present literature will contribute to the perception of the child and the design of the primary school buildings.

Child cognitive development and spatial perception

Childhood constitutes an important stage in the development of both physical and cognitive growth of the individual, and this period depends on many different genetic and environmental factors. The child interacts with the physical environment and uses the physical environment as a space for wondering, discovering, inventing and experiencing. The formation of child development also occurs with this dynamic action and the child's feedback (Moore, 1987). In order to comprehend the development of the child, it should be considered as a whole. It is necessary to know the physiological developments affecting the behavior of the child both directly and indirectly (Yavuzer, 2000). The socio-emotional development of the child in relation to his/her physical development is related to variables such as self-confidence, safety and comfort, interaction with the environment and definition of gender (Gur, 2014).

While Piaget and Inhelder (1967) expressed cognitive development in four stages as a sensory-motor period, a pre-operational period, a concrete operational period and a formal operational period, Vygotsky (1986) developed a theory that sees knowledge as built-in and cooperative. Vygotsky argued that knowledge diffuses between people and their environment, and that knowledge can be acquired by engaging with others through auxiliary activities (Gray *et al.*, 2012). However, according to Vygotskyan's theories, it is stated that the child's mental development and perception process are directly related to the environment in which they interact (Kozulin, 2003). The child communicates with the environment and reveals this by his/her behavior. It is known that factors such as genetics, family structure, culture and origin as well as physical environment have an impact on the behavior of the child. The interaction with the environment is common to every child, but the nature of the interaction is directly related to the opportunities provided by the environment (Olds, 1979).

In the context of cognitive development, it is important that there is consistency between child development, behavior and perception levels. Parallel development is continuous in adolescence. According to Piaget, the process in the 7–12 age group is referred to as the process during which knowledge is acquired through words and concepts as well as concrete processes. In this period, the child begins to think logically, but this thought is still limited to his/her concrete experiences (Pulaski, 1971). Visual reality begins to emerge and in the drawings both dimensional and proportional relations occur close to reality. Two-dimensional relations can be established (Clarke-Stewart and Parke, 2014; Gür *et al.*, 1989; Perkins, 2002). The child begins to perceive the environment more consciously with the physical conditions. In children, the concepts of number, time, space, size, volume and distance begin to settle.

As the perception of space cannot be expected from the child who has not developed a space concept, it cannot be expected to give a cognitive comment on space from the child whose bodily awareness has not emerged (Piaget and Inhelder, 1967). The child needs cognitive development and conceptual reinforcement for the formation of spatial perception (Vygotsky, 1986). In other words, the perception of space in the child is related to the space, the degree to which this perception influences the subconscious and how it shapes it. The perception of the built environment is directly related to the social-cultural development, physical development and psychological factors as well as the characteristics of the built environment.

Perception of the environment is carried out with the help of stimuli in the environment. While spaces containing variability can be perceived more, it is difficult to perceive the spaces that are ordinary. Since the formation of the built environment cannot be a result of a random setup, there is an order and organization. This is defined as shape-ground relations, completion, continuity, similarity and closeness by Gestalt principles that are one of the perceptual psychology studies (Rock and Palmer, 1990).

The method used to express perception in children may vary depending on age (Figure 1). The child begins to establish topological relations and projective relationships when he/she is in an egocentric spatial understanding between the ages of two and seven years. The child can establish spatial relationships by communicating more through the game. However, children who grasp projections at 7–11 years of age can comprehend

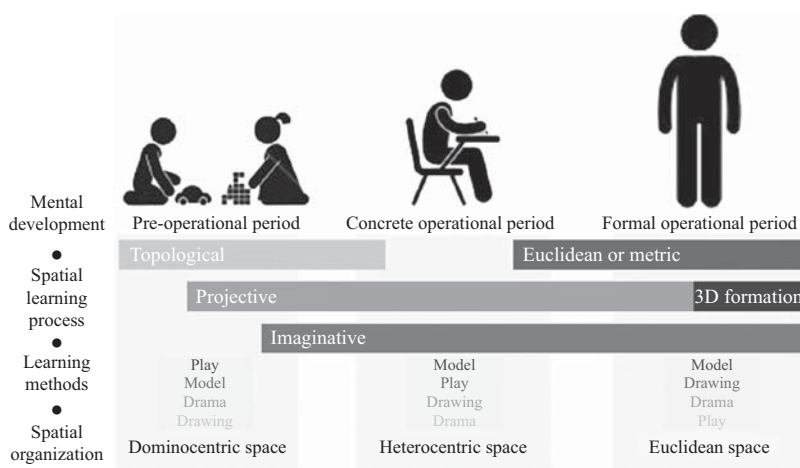


Figure 1.
Children's cognitive
development and
spatial perception

Sources: Taşçı (2014) and Türel (2017)

two- and three-dimensional expression methods. At this age, spatial relations are seen as mixed centered. After 12 years of age, the child begins to have metric space relations. The child begins to understand the Euclidean relations and the three-dimensional expression technique.

The concept of space in topological space relations is based on perspective and Euclidean relations (Akarsu, 1984). Every child undergoes a process of perception, but changes can be observed in processes, depending on variables such as environmental and hereditary. While the child grasps the space more intuitively until reaching the age of seven, he/she moves on to a new system that can be converted during the concrete processes. Akarsu (1984) defined this system as partitions, partitive addition, linear or cyclic alignment and symmetrical relationships.

Projective space relations are seen as the relation of objects or spaces with other objects and spaces in a perspective that includes the features of the environment in which an object is located. In this direction, the child can comprehend simple perspective relations. Toward the end of the concrete operations, the child can position the object in a spatial and spatial pattern relative to the reference point of the objects placed around a reference point, and a more accurate perspective schemes can be seen (Piaget and Inhelder, 1967).

Euclidean or metric space relations include topological relations and show a parallelism with the projective space. In the Euclidean space, the child forms the size of the objects, their relation to each other and their position in the space within a general framework of reference. Toward the end of the concrete transaction period (after 12 years of age), the child creates a turning point and develops a mapping system by establishing better relations between the spaces (Lynch, 1960). At the end of this period, the child begins to express his/her drawings within the framework of the correct perspective rules.

In the context of imaginative space, the components in the perception of the city are paths, borders, districts, nodes and landmarks. Paths are encountered in the form of a continuum and a functional requirement, such as streets, channels and pedestrian paths within the physical environment. Borders have continuity like roads. However, the roads are not binders but separators. Districts have an easily recognizable character. It can also be used as an external reference if the district is perceived from the outside. Nodes are strategic points that allow the user to enter a place and create routes. It can also be defined as the intersection points of routes. Finally, the landmarks that are easily identifiable as a reference point create a reference for many details about the environment (Lynch, 1960).

Primary school as a physical environment and child's spatial perception

The school building is the microcosmic model that best reflects the values and world views of the societies and their educational understanding (Öymen Gür and Zorlu, 2002). Hertzberger (2008) defines the school as a micro-city and the city as a macro-sized school. In the design phase of this built environment, it is thought that the development of production models that will keep up with the changes in educational areas rather than architectural concerns will affect the education of the child as well as his/her physical, emotional, social and cognitive development. In the future, the opportunities offered by the school will be more intertwined with the community and will enable students to be more secure and protective socially (Dudek, 2007). A primary school building in the city may appear as a building where only kids are educated, but it is actually a symbol of the city.

Educational structures significantly affect students' learning abilities, aspirations and motivation (Cole, 2011; Fisher and Dovey, 2016). In this context, the school enables the child to create different interactions in order to understand the child's own self and to offer a social environment for interpersonal relations (Levin, 1980; Rubin, 1998; Gur, 2014). It is thought that the spatial experience of the child is important, as the educational buildings are the foremost places where this process develops. Dudek (2000) argues that school

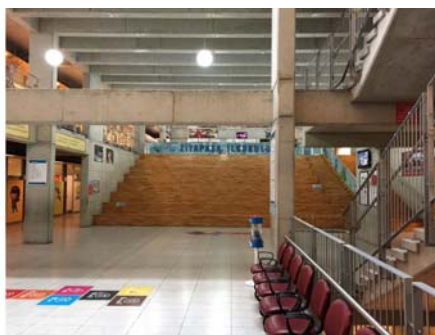
architecture should incrementally respond to social transformations in stages. However, it may be a challenge to design the school building for use in multiple user groups (Penoyre *et al.*, 2007). But it provides a mutual benefit in the urban-urbanite-child triangle in terms of sharing opportunities.

It is possible to say that the differences in the plan schemes of school buildings have an impact on the child. The spatial organization of modern school buildings has an open plan and flexible structure (Dudek, 2007; Mulcahy *et al.*, 2015; Gislason, 2007). According to Dudek (2007), modern school plans include a street plan, a campus plan and a mixed plan type. Street-planned school typology consists of a main circulation area and its connected circulation areas. They are economically feasible and suitable for future planning and change. In the campus-planned typology, the school is a set of semi-autonomous buildings, divided into specific areas. Each building contains different levels of classes, and each building has its own construct except for common areas such as an auditorium, a sports area and a cafeteria. The third plan type is the synthesis of previous plan types. The classes formed around a central courtyard are partly divided into levels (Dudek, 2007). Apart from these three plan types, there are also linear plan type schools produced due to easy and economical production. However, it is considered as a restrictive system because it cannot adapt to changes in the education system. Educational buildings in Turkey, as in many other countries, are designed with the cellular system. In other words, it consists of corridors that connect the learning areas in a box form.

Earthman and Lemasters (1996) investigated the relationship between a school's physical environment and children's behavior, development and success and found a link between these variables. Therefore, it is emphasized by many researchers of the effect of school design on a child's perceptual, physical, social-emotional development and learning (Şener, 2001). The flexibility and adaptability of school design symbolizes a school's way of thinking in the field of education, its relevance for psychological and pedagogical imperatives, its care and self-sacrifice for raising consciousness, and the questioning of generations. There are also various parameters that affect the users of the school building as a platform for learning, entertainment, social and cultural sharing for both the children and the locals who are not used for education (Türel, 2017).

Case study: the effects of school environment's spatial and physical characteristics on children's spatial perception in Kagithane

A case study was carried out in three primary schools with different typologies in the Kagithane district of İstanbul. The first is the Ziyapasa Primary School (School 1) located in the Ortabayir neighborhood of the Kagithane district. School 1 is one of the state schools designed by Uygur Architecture and has a unique design (Plate 1). The second school



Source: Türel (2017)

Plate 1.
Exterior and
interior perspective
of School 1

selected for the case study is the Nef Primary School (School 2) located in the Ortabayir neighborhood of the Kagithane district. School 2 is one of the state schools design by Cinici Architecture and has a unique design (Plate 2). The last one is the Ismail Erez Primary School (School 3) located in the Hamidiye neighborhood of the Kagithane district. The school is one of the type project schools designed with the same plan typology and building character by the Ministry of National Education (Plate 3). These schools were to be prepared for a possible earthquake in Istanbul and to be reconstructed in the context of the “Istanbul Seismic Risk Reduction and Emergency Preparedness Project.” The case study was carried out in an environment with similar characteristics in a single region in order to stabilize the education and income level of the parents. These variables would cause an increase in the reality and quality of children’s drawings. Attention was also paid to the fact that the student capacities and the physical sizes of the schools were close to each other. The case study was performed with the third and fourth grade students in order to obtain more realistic and reliable data.

Methodology

The study was carried out with the 9–10 age group that represents the period of tangible operations. Qualitative and quantitative research methods were used together in the case study. Observations in the case study were performed three times in two weeks at all three schools. At least one of the observations was made in bad weather conditions. Thus, it was

Plate 2.
Exterior and interior
perspective of
School 2



Source: Turel (2017)

Plate 3.
Exterior and interior
perspective of
School 3



Source: Turel (2017)

tried to observe the behavior of children when they had to use interior spaces of the school. Existing cameras in schools supported the observations that were made for a 40-min lunch break in 10-min periods. The number of students in the observed schools is 1,200 in School 1, 1,100 in School 2 and 1,000 in School 3. During this time, data were collected by observing where the children spend their time, which physical activities they prefer, the difficulties they encounter as a result of their spatial experiences and the solutions they found.

The drawing study was carried out with the participation of 172 students in the schools of Ziyapasa, Nef and Ismail Erez, and children were asked to sketch the spaces they remembered in the school. With this study, it was tried to discover how children perceive the school building in their subconscious based on their own observations and past experiences. Topological, projective, metric and imaginative space parameters were taken into consideration in order to measure the spatial perception levels of children from the physical environment and to evaluate cognitive maps. These maps were evaluated according to the existence of parameters. The cognitive maps, which are drawn together with both the interior and the close environment of each of the parameters, are examined according to the following variables: topological space; projective space; metric space; and imaginary space.

Results of the observations

Student behavior in the interior and exterior places of the school was observed during a 40-min lunch break in 10-min periods in all three schools. The frequency of usage of the spaces used by the students and the purpose of use were also considered as variables. Observations were made on three different days in each school, with at least one day being in bad weather conditions. In the observations, some situations and behavior were considered, such as children's activities in interior and outdoor spaces of the schools, as individuals or groups, and movement directions and nodes. In order to determine how and for what purpose they used the school space, they were categorized as mobile and immobile. Mobile and immobile users are shown together for comparison with each other. The observation findings of all three schools are given below.

School 1. It was observed that the corridors were preferred as a socialization environment by the students except in the open and free open spaces created on each floor when the outdoor space could not be used (Figure 2). Although the building was designed for 1,000 people, it is seen that 1,506 students are currently educated in the school, and the students used the school's interiors frequently during the break and lunch periods. The inadequacy of the schoolyard causes children to produce solutions in the interior spaces for outdoor activities. It was observed that children preferred the colored areas with different patterns and more sheltered sections.

Each floor of the building provides a spatial organization that can be perceptible and traced through gallery gaps. Therefore, the children can meet their physical and spatial needs actively not only on their own floors, but also on other floors. It was seen that children choose the appropriate parts of the space according to the games and activities for small or large groups, and they organize and use the space according to their needs. However, it was observed that many of the students spend time in other parts of the school rather than in the classroom. It was observed that a space concept emerged in front of each class. As they moved away from the classes, the students played in more crowded groups and spent time with their friends. It was generally seen that there are crowded groups in the corridors where there are no classrooms that are not overwhelming in size, and away from gallery space.

In general, the effect of weather conditions has been found to be less effective in interior usage. As a result of the observations made under bad weather conditions, the effect of weather on interior and outdoor usage of the school was found to be low. It was seen that the children spent their time in interior and outdoor areas rather than in their own classes.

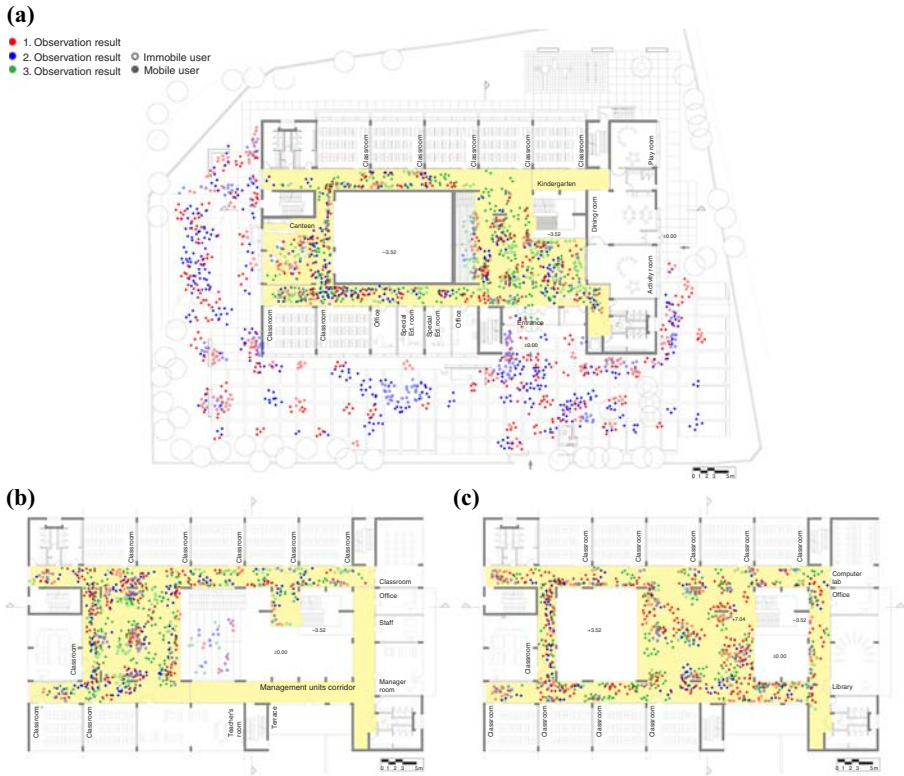


Figure 2.
School 1 entrance
floor (a), 1st floor (b)
and 2nd floor (c)
observation findings

It was observed that the interior space was used as an active playground-socialization space, and the children were very active on the vertical axes.

School 2. Although the school has a capacity of 970 students, it was seen that both the indoor and outdoor activity areas and free spaces were insufficient. Therefore, the children often preferred to stay in the classrooms or use the schoolyard depending on the weather conditions (Figure 3). It was seen that children use the free spaces on the ground floor actively for their activities. The physical and spatial facilities of School 1 and the presence of the dining hall of School 3 provide a place for children to satisfy their eating and drinking needs. However, it was determined that needs in School 2 were mostly met in the classroom.

It was observed that children often carried out activities in their classrooms or corridors because there are few free or designed spaces in the interior space of the building where large groups could perform activities together. For this reason, socialization with other students often took place in the eating-drinking area and the schoolyard. The parts with various studios were used more frequently for games and activities compared to the parts of classes. It is thought the parts caused it with various studios being more isolated and away from the vertical circulation areas.

School 3. It was observed, notwithstanding the weather conditions, that the majority of the children preferred to spend their breaks in the schoolyard (Figure 4). Also, if they had limited time, they preferred to be in their classroom. In this school, with a large schoolyard compared to the other schools, children used the schoolyard for all kinds of activities. As to interior space, it was seen that the corridors were only used to reach the vertical



Figure 3. School 2 entrance floor (a), 1st floor (b) and 2nd floor (c) observation findings

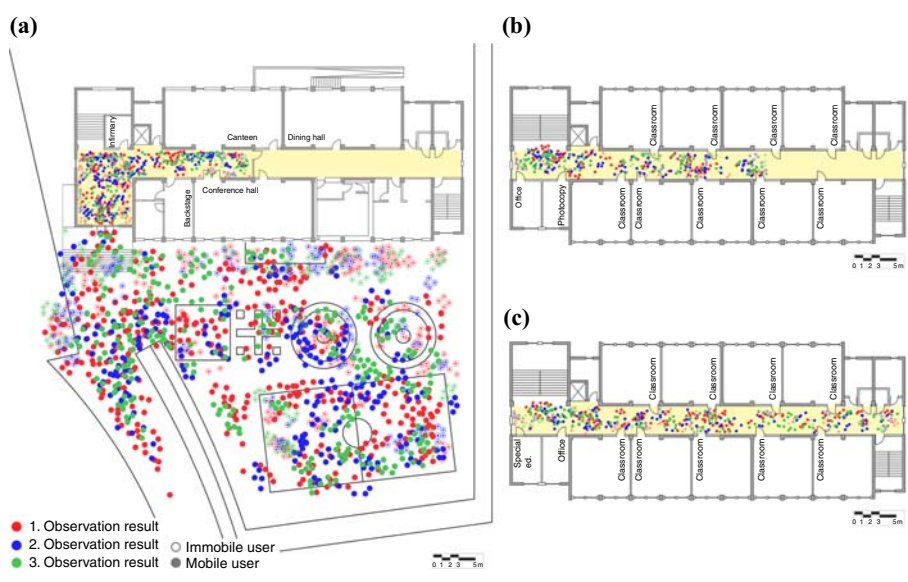


Figure 4. School 2 entrance floor (a), 1st floor (b) and 2nd floor (c) observation findings

circulation area. They only used a part of the corridor for the short-term games and meeting with friends. This part of the corridor is lighter and not as intense as the fronts of the classes.

As the relations between the floors were not strong, it was observed that children had less opportunity to socialize, and children on the third and fourth floors preferred to be in their classrooms during breaks. It was also seen that the opening of the doors of the classrooms directly into the corridors and the lack of light in the common spaces caused some accidents. In addition, the corridors that were located on both the first floor where the kindergarten is located and the ground floor where the dining hall is located, and where one-sided light was received and the corridor area was narrowed. For this reason, the children spent less time in these areas and quickly went to the garden.

Results of the cognitive maps

In the cognitive mapping study, selected children from grades 3 and 4 from each school were asked to draw their school building from the inside and outside. They were given 15 min for each drawing. The distribution of participants by schools is shown in Table 1. Topological, projective, metric and imaginative space parameters were used in order to measure the children's spatial perception levels and to evaluate cognitive maps. Cognitive maps were evaluated according to the existence of parameters.

The following variables were evaluated both as interior and exterior spaces according to the presence of districts, edges, paths, nodes and landmark: topological space, proximity, disconnection, continuity, layout and similarity; projective space, perspective, straight line, parallel line drawing status; metric space, proportionality, location, conservation parameters; and imaginary space. In addition, in the drawings, the presence of references such as reinforcement, material and color used when drawing interior and exterior spaces were also examined. Based on the drawings of the children participating in the cognitive mapping study, it was determined that there were some basic differences between the three schools in terms of their spatial perception.

Topological space properties. In all three schools, it was seen that the topological space features appeared in the drawings of most students (Table II) (Figure 5). Approximately three quarters of the students participating in the research completed the concrete operational period and entered into the period of abstract operations. It is seen that the students in School 1 show more topological space properties than other schools. The emergence of this situation is thought to change in direct proportion with the school having more visual stimuli and stimulants.

Table I.
Distribution of
students participating
in the study according
to schools

Number of students participating in the study	Girls	%	Boys	%	Total
Ziyapasa PS (Sch 1)	30	48	32	52	62
Nef PS (Sch 2)	31	55	25	45	56
Ismail Erez PS (Sch 3)	33	61	21	39	54
Total	94		78		172

Table II.
Comparison of
topological space
properties observed
in students' drawings
by schools

Topological space	Exists	%
Ziyapasa PS (Sch 1)	53	85
Nef PS (Sch 2)	39	70
Ismail Erez PS (Sch 3)	41	76



Notes: (a) School 1, 10 years old; (b) School 1, 10 years old

Figure 5.
Drawings expressing
the topological
space properties

Projective space properties. In the projective space properties, it was observed that children had the highest level of ability to draw straight and parallel lines in all three schools. As expected, it is a common result in children who finish the period of concrete operations. However, in its simplest form, it is understood that the ability to draw in perspective is higher in Schools 1 and 2 (Table III) (Figure 6).

In this age group, it is noteworthy that the ability to express the locations of newly formed objects in perspective is more in Schools 1 and 2 than in School 3. This is related to the physical and spatial structure of the school. Therefore, it can be stated that children who study in school buildings designed with context-specific data are more competent in three-dimensional expression than children studying in type projects.

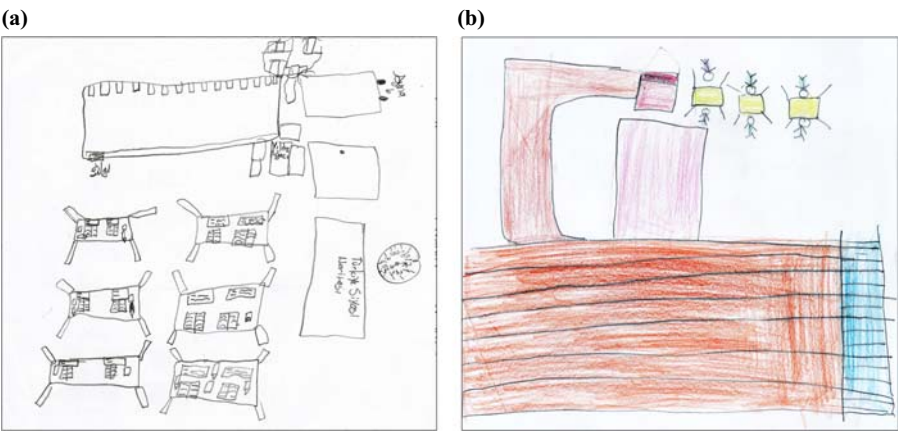
Metric space properties. Metric space properties are predicted to start at the age of ten. The metric space is expressed as the physical environment and the way in which objects are drawn with correct perspective rules. Drawing studies are detailed in this context as conservation and block expression. Conservation is defined as the child's drawing of the objects with his/her true proportions in the framework of perspective rules. Block expression is defined as the correct perspective rules according to the escape points of the object or space.

In metric space properties, three students in School 1 have a conservation concept and only one student from the other schools. This may be interpreted as the fact that metric

Projective space properties	Exists	%
<i>Ziyapasa PS (Sch 1)</i>		
Straight line	62	100
Parallel line	62	100
Perspective	8	13
<i>Nef PS (Sch 2)</i>		
Straight line	56	100
Parallel line	56	100
Perspective	9	16
<i>Ismail Erez PS (Sch 3)</i>		
Straight line	52	96
Parallel line	52	96
Perspective	3	6

Table III.
Comparison of
projective space
properties observed
in students' drawings
by schools

Figure 6.
Drawings expressing
the topological space
properties



Notes: (a) School 3, 8 years old; (b) School 1, 9 years old

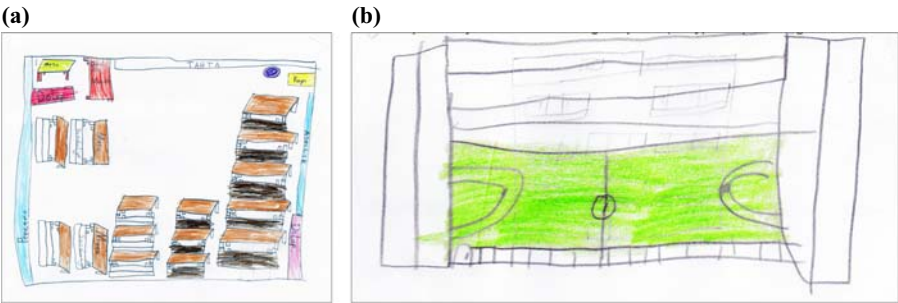
space properties are not fully developed in the 9–10 age group. The distribution of the drawings showing metric space properties according to schools is shown in Table IV (Figure 7).

Imaginative space properties. It is seen that the interior and exterior spaces of the school buildings have different levels in the drawings of children. In the first phase of the drawing,

Table IV.
Comparison of metric
space properties
observed in students'
drawings by schools

Metric space properties	Draw	%
<i>Ziyapasa PS (Sch 1)</i>		
Block expression	8	13
Conservation	3	5
<i>Nef PS (Sch 2)</i>		
Block expression	9	16
Conservation	1	2
<i>Ismail Erez PS (Sch 3)</i>		
Block expression	8	15
Conservation	1	2

Figure 7.
Drawings expressing
the metric space
properties



Notes: (a) School 2, 8 years old; (b) School 2, 9 years old

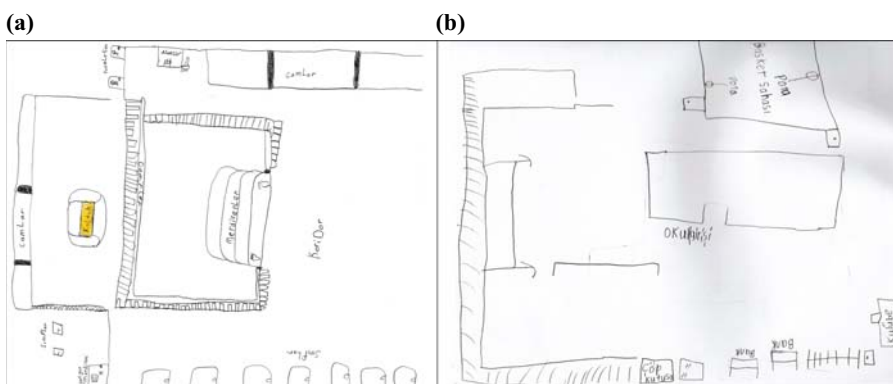
children were asked to draw the inside of the school building within 15 minutes. In the second stage, again in 15 minutes, they were asked to draw the school building and its surroundings, as they perceived it from outside.

Results at the first drawing stage. In the first stage of the drawing, the imaginary space properties in the drawings of the children are shown in Table V. According to the table, it is revealed that there are more imaginative space properties in School 1 than in the other schools (Figure 8). This situation can be interpreted that the children in School 1 have more experience in the inner space of the school, and that the school has rich stimuli.

Borders, which are the most frequently encountered parameters in the physical environment of children, have not been seen in the children's drawings as expected. A similar situation appears in the image of the district parameters. This situation can be interpreted as the fact that the imaginary elements are not obvious in the places where children live in the

Table V.
Comparison of
imaginative space
properties observed in
students' drawings of
the inside of building
by school

Projective space properties	Draw	%
<i>Ziyapasa PS (Sch 1)</i>		
Paths	25	40
Districts	44	71
Borders	27	44
Nodes	36	58
Landmarks	35	56
<i>Nef PS (Sch 2)</i>		
Paths	12	21
Districts	26	46
Borders	9	16
Nodes	23	41
Landmarks	23	41
<i>Ismail Erez PS (Sch 3)</i>		
Paths	17	31
Districts	29	54
Borders	19	35
Nodes	26	48
Landmarks	28	52



Notes: (a) Inside of the buildings, School 1, male, 10 years old; (b) outside of the buildings, School 1, male, 9 years old

Figure 8.
Drawings expressing
the imaginative space
properties

physical environment. However, in the drawings of students in School 1, there are more distinct parameters than other schools. The same applies to paths and nodes parameters.

Significant landmarks drawn in the interior space by children within the context of the landmarks parameter, which is one of the imaginative space properties, are given in Table VI. In the interior drawings of the children who participated in the case study, there are class landmarks such as table, desk, cupboard, teacher's desk and blackboard. Classrooms, which are the most prominent spaces in the children's drawings, are also the space where the children give the most details. Playgrounds were seen in the drawings of the students in all three schools.

Results at the second drawing stage. In the second phase of the drawing, it was seen that the imaginative properties of the children in Schools 3 and 1 were higher than those in School 2 (Table VII). It can be said that the children in Schools 1 and 3 have more imaginative perceptions of the school's garden and its surroundings. The students in School 1 are found to have more spatial perceptions about both interior and exterior of the school buildings (Figure 8). However, it can be said that children in School 3 have more spatial perceptions about outdoor space as they spend more time in the schoolyard during the break and lunch periods.

Table VI.
Comparison of the obvious landmarks observed in the interior drawings of the students by schools

Drawn landmarks	Ziyapasa PS (Sch 1)	Nef PS (Sch 2)	Ismail Erez PS (Sch 3)
Table	18	27	28
Desk	16	27	15
Cupboard	17	20	13
Teacher's desk	4	17	21
Blackboard	4	15	16
Playground equipment	24	17	21
Colors	3	1	0
Texture	1	2	0
Elevator	10	7	5
Others	1	8	0

Table VII.
Comparison of imaginative space properties observed in students' drawings in the outside of building by schools

Imaginative space properties	Draw	%
<i>Ziyapasa PS (Sch 1)</i>		
Paths	17	27
Districts	39	63
Borders	17	27
Nodes	26	42
Landmarks	27	44
<i>Nef PS (Sch 2)</i>		
Paths	7	13
Districts	17	30
Borders	7	13
Nodes	14	25
Landmarks	13	23
<i>Ismail Erez PS (Sch 3)</i>		
Paths	9	17
Districts	34	63
Borders	15	28
Nodes	29	54
Landmarks	31	57

It is clear that the images of the districts, borders, nodes and landmarks are more in the drawings of children in Schools 1 and 3 than in the School 2. However, path images were seen in the drawings of the children in School 1 more than in the other schools. The path image is the least common parameter in the drawings of the majority of children in the second drawing study as it also was in the first drawing study.

In the second drawings of the students, the frequency of outdoor usage and the details show that the relationships were significant. It is also observed from the previous data that the students in School 1 have strong perceptions about the whole school. This situation is supported by outdoor drawings (Table VIII). On the other hand, it was supported by the observations that the children in School 3 used the outdoor space rather than the interior space. The data from the drawings also support this situation. Due to the richness of the stimuli in the outdoor area, it is thought that children in Schools 3 and 1 use more reference points in their drawings.

General evaluation of the results

Findings from children's drawings and observations show that children have higher topological, projective and metric space properties in schools designed with rich stimulants and specific data. It has been determined that children are reflected by the cognitive maps of the districts they use in indoor and outdoor spaces. Another detail in the drawings is the importance of reference points in buildings. There are many reference points in School 1, while it is revealed that the lack of reference points in Schools 2 and 3 is reflected in the drawings of the children. It was observed that the perception of space varied as to classroom and building scale. In Schools 2 and 3, while the participants mostly draw only their classrooms, in School 1, there is a perception not only of their classroom but also of the whole school (Figure 9). Students' use of spaces outside the classroom, such as circulation and service spaces, revealed the influence of spatial perception.

A similar difference was observed in the perception of the students as a whole with the surroundings of the building (schoolyard, building shell, roads, focal points, etc.). It is seen that the building shell and its environments were drawn in more detail at School 1, whereas in the drawings of School 3, only the schoolyard was detailed (Figure 10). Due to the lack of specialized details to the school, the details of the building shell and its surroundings were found to be fewer in the drawings of the children in Schools 3 and 2. Furthermore, places such as teacher's room, dining hall, gymnasium, conference hall and library, which are less frequently located in children's drawings, are not places frequently used by children. Therefore, these spaces are not sufficiently involved in the spatial perceptions of children.

It was observed that the results obtained from the cognitive mapping study coincided with the findings obtained from the observations. It has been supported by observations that children frequently used the spaces in the cognitive mapping study for some activities. In all three schools, it was seen that children used different, flexible, and adaptable and more sheltered spaces to play, chat, meet friends. Therefore, it can be said that the children's who are educated in flexible, free and adaptable spaces has an effect on both their behavior and their spatial perception.

Drawn landmarks	Ziyapasa PS (Sch 1)	Nef PS (Sch 2)	Ismail Erez PS (Sch 33)
Schoolyard borders	17	0	4
Tree	5	1	22
Bench	10	5	18
Bin	6	0	0
Playground equipment	2	3	15
Building details	16	0	0

Table VIII.
Comparison of the
obvious landmarks
observed in the
exterior drawings of
the students by
schools

Figure 9.
The comparison of
the spaces observed
in the interior
drawings of the
students by schools

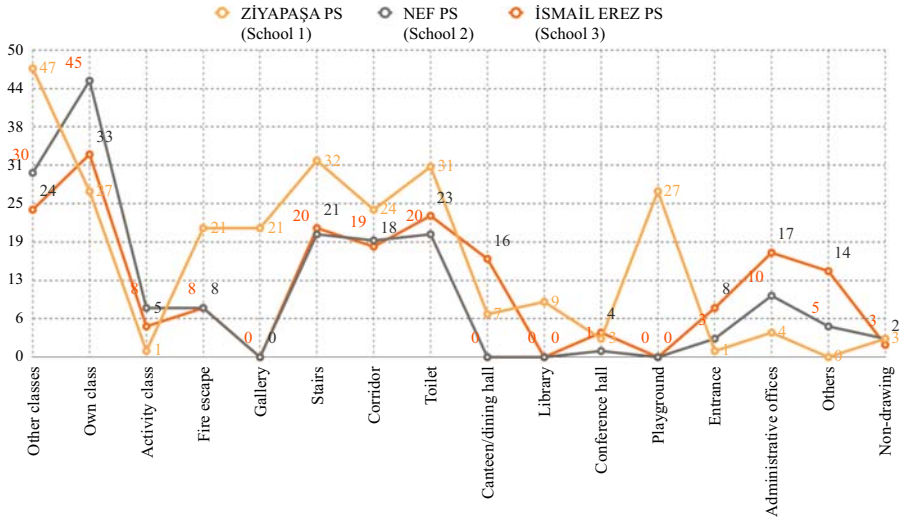
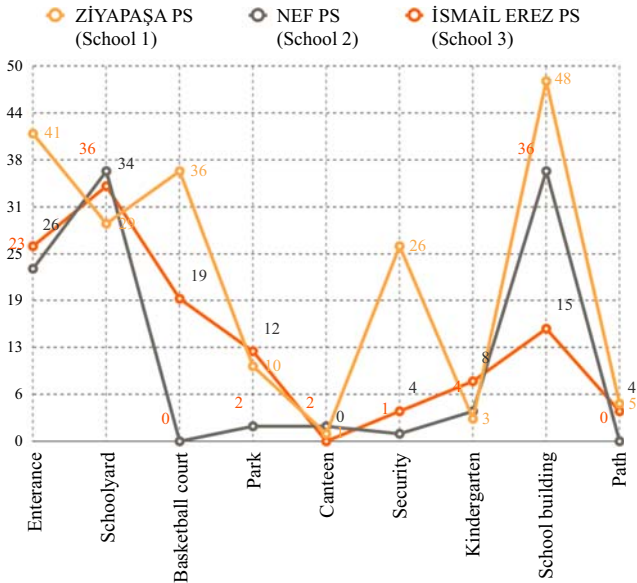


Figure 10.
The comparison of
the spaces observed
in the exterior
drawings of the
students by schools



Conclusion

The study aiming to investigate the effects of spatial and physical characteristics of primary school typologies on the spatial perception and behavior of the child was conducted with children aged between nine and ten years. The visual reality, Euclidean and two-dimensional relations begin to form in this children's age group, and the objects are expected to be more accurately expressed both dimensionally and proportionally in their drawings (Gür *et al.*, 1989; Perkins, 2002; Piaget and Inhelder, 1967). Therefore, it can be said that it also carries a transition from the period of concrete operations to the period of formal operations. It was observed that these transitions were not fully completed in all schools in different typologies.

It is interpreted that the perspective perception in the drawings of children is generally weak, and metric space properties are not sufficiently covered in their cognitive maps. Attention was paid as to whether topological, projective, metric and imaginative space properties were used in the evaluation of cognitive maps.

When looking at the cognitive map analysis in general, the spaces in the children's cognitions in the primary school building spatial organization can be seen; it has also been seen that they have high social interaction, no sharp borders, free and internalized areas. In socially interactive spaces, communication of people with other people and the physical environment allows for a definition of space and meaning of the environment (Sanoff, 1994). In the drawing study, it was seen that children give more place to such spaces. In this context, the physical features and spatial organization of the buildings contribute to the development of the child's spatial perception (Şener, 2001). It is observed that the formation of visual stimuli (such as color, texture, shape) that helps buildings gain identity by the Lynch's (1960), in buildings, interior spaces and positively the building envelope designed by taking into consideration context-specific data, affect the spatial perception in children of nine to ten years of age in the period of concrete operations. As another method of the case study, the use of space and the behavior of space obtained from observations support this phenomenon.

The perception status is affected by a single place on a wide scale, and this result is supported by the case study. In the Nef and Ismail Erez Primary Schools, the frequency of children who just drew their own classroom revealed that children's perceptions of the whole school are weaker. Also, a similar situation has been observed in which many children spend time in their own classrooms. Although the Nef Primary School, like the Ziyapasa Primary School, was designed with context-specific data, it was found that the level of perception of children in the Ziyapasa Primary School had a higher level of perception about the whole school and its surroundings, thanks to its location in the city and its interiors that allow visual communication. In addition, it can be said by the study that the creation of the spaces where students can build social relations in the physical environment of the building are able to develop their creativity, explore, gain different experiences and reconstruct them so as to strengthen the spatial perception.

In conclusion, primary education is one of the factors that plays an important role in the cognitive development and social relations of the individual. Education is not only a process but also an open and changing transforming system that communicates with its environment (Hertzberger, 2008). Therefore, the education system, which is designed to be able to keep up with these changes and transformations, can only be provided by educational buildings with spatial organization that can adapt to the innovations in parallel. The relationship between the child and the physical environment is basically formed by the desire to establish a bond. It is important that the physical environment of the primary school be well constructed and functional because the child discovers, tests, imitates and actively learns the environment. Therefore, there should be a number of free opportunities for the physical environment of the primary school. It is important to create social and flexible spaces where the child can discover, experience, and develop creativity on his/her own. When designing the primary school building, it should be in a spatial organization full of rich stimuli, allowing visual communication, and containing unfinished areas to explore. Linear, introverted and closed (inelastic) spatial constructs should be avoided. The importance of the organization of the circulation areas, which enables direct communication with other people during breaks and lunch periods, is clearly seen. Therefore, circulation areas are often used as socializing spaces with other people. These areas should also support this socialization and allow children to easily explore their surroundings. It is important that transitions between spaces should also provide sustainability within this framework.

Within the scope of the study, state primary schools of the Ministry of National Education were examined. It is thought that the study will contribute to future school

building design in a way to raise awareness of children's physical, social and perceptual development. The study will contribute to the enlightenment of primary school design studies, and will give opportunity to the newly designed primary school buildings' architects to realize their design in order to improve the child's spatial perception and to support his/her spatial experience. The study can be used by the Ministry of National Education, municipalities, architects, and can be a base for designing further primary school building typologies.

References

- Akarsu, F. (1984), "Piaget'e göre çocukta mekan kavramının gelişimi (According to Piaget, development of space concept in child)", *Mimarlık*, Vol. 84 No. 207, pp. 31-33.
- Clarke-Stewart, A. and Parke, D. (2014), *Social Development*, Wiley Global Education, NJ.
- Cole, A. (2011), *Critical Review of Elementary School Design*, unpublished master thesis, University of Massachusetts Amherst, MA.
- Dudek, M. (2000), *Architecture of Schools: The New Learning Environments*, Routledge, New York, NY.
- Dudek, M. (2007), *Schools and Kindergartens: A Design Manual*, Walter de Gruyter, Berlin.
- Earthman, G. and Lemasters, L. (1996), "Review of research on the relationship between school buildings, student achievement, and student behavior", paper presented at the Annual Meeting of the Council of Educational Facilities Planners, International, Tarpon Springs, FL, October 8, Council of Educational Facility Planners, International, Scottsdale, AZ.
- Fisher, K. and Dovey, K. (2016), "Plans and pedagogies, school design as socio-spatial assemblage", in Fisher, K. (Ed.), *The Translational Design of Schools: Advances in Learning Environments Research*, Vol. 7, SensePublishers, Rotterdam, pp. 159-178.
- Gislason, N. (2007), "Placing education: the school as architectural space", *Philosophical Inquiry in Education*, Vol. 16 No. 3, pp. 5-14.
- Gray, S.L., Culpepper, C.L. and Welsh, D.P. (2012), "Adolescence", in Ramachandran, V.S. (Ed.), *Encyclopedia of Human Behavior*, 2nd ed., Elsevier Science, London, pp. 22-27.
- Gur, E. (2014), "The effect of physical and environmental factors of a 'child development center' on a center's selection", *International Journal of Architectural Research: ArchNet-IJAR*, Vol. 8 No. 3, pp. 136-148.
- Gür, Ş., Özbilen, A. and Ertürk, S. (1989), "Çevresel psikolojide çocuk-ilkokul öğrencilerinin okul çevresine bilişsel değerlendirmelerinin tasarım kararlarına etkisi (The effect of cognitive evaluations of primary school students on school environment on design decisions)", Karadeniz Teknik Üniversitesi, Mühendislik Mimarlık Fakültesi Genel Yayın, Trabzon.
- Hertzberger, H. (2008), "Space and learning: lessons in architecture 3", nai010 Publishers, Rotterdam.
- Kozulin, A. (2003), *Vygotsky's Educational Theory in Cultural Context*, Cambridge University Press, Cambridge.
- Levin, S. (1980), *The Child's Conception of Space and Painting: Some of Piaget's Ideas Related to a Cezanne Water-Colour*, unpublished master thesis, Concordia University, Montreal.
- Lynch, K. (1960), *The Image of the City*, MIT Press, Cambridge, MA.
- Moore, G.T. (1987), "The physical environment and cognitive development in child-care centers", in Weinstein, C.S. et al. (Eds), *Spaces for Children*, Plenum Press, New York, NY, pp. 41-72.
- Mulcahy, D., Cleveland, B. and Aberton, H. (2015), "Learning spaces and pedagogic change: envisioned, enacted and experienced", *Pedagogy, Culture & Society*, Vol. 23 No. 4, pp. 575-595.
- Olds, A.R. (1979), "Designing developmentally optimal classrooms for children with special needs", in Meisels, S. (Ed.), *Special Education and Development: Perspectives on Young Children with Special Needs*, University Park Press, Baltimore, MD, pp. 91-138.
- Öymen Gür, Ş. and Zorlu, T. (2002), *Çocuk mekanları (Children's spaces)*, YEM Yayınları, İstanbul.

- Penoyre, L., Penoyre, G. and Prasad, S. (2007), *Transformations: The Architecture of Penoyre & Prasad*, Black Dog Publishing, London.
- Perkins, B. (2002), *Building Type Basics for Elementary and Secondary Schools*, John Wiley & Sons, NJ.
- Piaget, J. and Inhelder, B. (1967), *The Child's Conception of Space*, WW Norton, New York, NY.
- Pulaski, M.A.S. (1971), *Understanding Piaget: An Introduction to Children's Cognitive Development*, Harper & Row Publishers, New York, NY, ERIC.
- Rock, I. and Palmer, S. (1990), "Gestalt psychology", *Scientific American*, Vol. 263 No. 6, pp. 84-90.
- Rubin, K.H. (1998), "Social and emotional development from a cultural perspective", *Developmental Psychology*, Vol. 34 No. 4, pp. 611-615.
- Sanoff, H. (1994), *School Design*, Taylor & Francis Publishers, ERIC, New York, NY.
- Şener, E.A. (2001), "Okul öncesi çocuk eğitim merkezi için değişebilir/dönüştürülebilir/esnek bir 'fiziksel çevre modeli' (A changeable/transformable/flexible 'physical environment model' for child development centers)", doctoral thesis, İstanbul Teknik Üniversitesi, İstanbul.
- Türel, A. (2017), "İlkokul fiziksel çevresinin çocuğun mekansal algısına ve davranışına etkisi: Kağıthane örneği (Effects of physical environment of primary school on children's spatial perception and behavior: the case of Kağıthane)" (Advisor: Assoc. Prof. Elmira Gür), unpublished master thesis, İstanbul Teknik Üniversitesi.
- Vygotsky, L.S. (1986), "Thought and language" (Newly revised and edited by Alex Kozulin), The Massachusetts Institute of Technology, MIT Press, Cambridge, MA.
- Yavuzer, H. (2000), *Çocuk psikolojisi (Child Psychology)*, Remzi Kitabevi, İstanbul.

Further reading

- Taşçı, B.G. (2014), "Çocuk mimarlık çalışmalarının değerlendirilmesi ve ilköğretim için yapısal çevre eğitim programı önerisi (Sosyal bilgiler dersi için) (Evaluation of children's architecture and environmental education program proposal for primary education: for social studies course)", doctoral thesis, Dokuz Eylül Üniversitesi Fen Bilimleri Enstitüsü, İzmir.

About the authors

Ahmet Türel completed his undergraduate education in 2013 at KTU, Department of Architecture. In 2014, he started to work as a research assistant in the Department of Architecture in BTU. He was assigned as Research Assistant at the Department of Architecture in ITU to complete his postgraduate studies. He has received his Master's Degree in Architectural Design in 2017, from ITU with a thesis entitled "Effects of Physical Environment of Primary School on Children's Spatial Perception and Behavior: The Case of Kağıthane." He has participated in several national and international workshops and get prizes several architectural design competitions. He still continues to participate in architectural design competitions. He is currently PhD Student and Research Assistant at ITU.

Dr Elmira Ayşe Gür (Associate Professor) has received her PhD in architectural design in 2001, from ITU with a thesis entitled "A Changeable/Transformable/Flexible "Physical Environment Model" for Child Development Centers." She has been Visiting Scholar in North Carolina State University (NCU) and Carnegie-Mellon University between 1998 and 1999. She has participated in several international workshops and conferences. She had prizes at several architectural design competitions and her studies and writings have been published nationally and internationally. Gur's research interests lie within a wide spectrum of areas from architecture to urban development. She has published and edited several papers and books on affordable housing, squatter settlements, housing development and typology in Istanbul, design studio's physical environment, design education, child development centers, post-disaster temporary shelters and urban transformation. Dr Elmira Gur is also Vice Dean at the Faculty of Architecture ITU. Dr Elmira Ayşe Gür is the corresponding author and can be contacted at: elmiragur@itu.edu.tr

For instructions on how to order reprints of this article, please visit our website:

www.emeraldgroupublishing.com/licensing/reprints.htm

Or contact us for further details: permissions@emeraldinsight.com